Service terms and description

NetApp Keystone

NetApp
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Flex Subscription service terms and descriptions

- NetApp Keystone Flex Subscription (Flex subscription) is available for a minimum of one year and up to three years. After the initial term, the service is renewable on an annual basis. Capacity can be increased in increments as small as 1 TiB.
- The minimum capacity is 100 TiB per site, and each site can have one or more clusters to meet the minimum capacity requirement. In a partner-operated model, subscriptions with flexible minimums are created for a customer, per site and across service levels.
- The 100 TiB capacity can be one single performance level or a combination of levels.
- Tenant subscriptions are limited to service levels that partners are subscribed to.
- 20% of burst capacity is available at each site; any burst usage is billed only for that billing period. If you need an additional burst requirements that is greater than 20%, contact support.
- Committed capacity or performance levels cannot be altered during a contract term.
- Increasing capacity or changing to higher performance level during term is allowed; however, moving from a higher performance level to a lower level is not permitted.
- Any change request in the last 90 days of the term requires the customer to renew the service for a minimum of one year.

Flex Subscription service capacity definitions

The NetApp Keystone Flex Subscription (Flex Subscription) service capacities include:

- **Logical capacity.** Data placed into the Flex Subscription infrastructure by a customer:
  - All Flex Subscription capacities refer to a logical capacity.

    For example, if a 1 TiB file is stored on the Flex Subscription infrastructure, then at least 1 TiB of capacity must be purchased.
- **Committed capacity.** Minimum logical capacity billed each month for the duration of the term:
  - Capacity is committed to each performance level.
  - Committed capacity cannot be decreased during the term.
  - Committed capacity and additional performance levels can be added during the term.
- **Burst capacity.** Logical capacity, in excess of committed capacity:
  - Flex Subscription services provides 20% capacity in excess of the committed capacity.
  - Burst capacity can be consumed on an elastic basis and is charged on a daily, average-consumed basis.
  - Burst capacity up to 20% is charged at a same rate as the committed capacity.
  - Burst capacity greater than 20% of committed is charged at a premium rate. Contact support for any additional burst requirements greater than 20%.
- **Consumed/provisioned capacity.** Sum of all configured volume sizes:
  - Capacity is provisioned through the creation, modification, deletion, or potential auto-growth of volumes.
  - Includes snapshots and clones.

- The amount of data stored within provisioned capacity is not considered.
- The amount of data actually written to disk is not considered.

- **Billed capacity.** Monthly bill = (committed capacity [TiB] * committed rate [$/TiB]) + (daily average provisioned burst capacity [TiB] * burst rate [$/TiB]):
  - The monthly bill contains a minimum charge based on the committed capacity.

    The monthly bill varies beyond the minimum charge based on daily average burst capacity consumption.

Performance Service Levels

NetApp Keystone Flex Subscription (Flex Subscription) offers capacity at predefined performance levels.

Each Performance Service Level is defined by its I/O density, which is the ratio of performance (input/output operations per second [IOPS]) and used storage (TiB of stored data) which is IOPS/TiB.

Each volume managed by the Flex Subscription services is associated with a performance service level. All I/O operations and all used storage used on the respective volume are factored into the volume's I/O density calculation.

The below figure defines the performance service levels.

	File and Block						Object	Block		
	Extreme	Extreme w/data tiering[2]	Premium	Premium w/data tiering[2]	Standard	Value	Object	Extreme	Premium	Standard
Workload type	Analytics, databases		VDI, virtualization apps, SW dev		File shares, web servers	Backup	Media repository, archiving	HPC	Video surveillance	Backup
Target IOPS/TiB	6,144	6,144[1]	2,048	2,048[1]	128	N/A	N/A	N/A	N/A	N/A
Max IOPS/TiB	12,288	12,288[1]	4,096	4,096[1]	512	N/A	N/A	5,500	4,000	N/A
Max throughput MBps (32KB/IOP)	384	384[1]	128	128[1]	16	N/A	N/A	43	31	N/A
Latency	<1 ms	<1 ms[1]	< 2 ms	< 2 ms[1]	<17 ms	N/A	N/A	< 0.5 ms	<0.5 ms	N/A
Minimum capacity	100TiB*^						1 PiB	100 TiB	100 TiB	300 TiB
Protocols	NFS, CIFS, iSCSI, FC						S3	FC, iSCSI		

* Minimum one-year term and 100TiB of storage.
^Channel led opportunities: Starts with 15TiB for Extreme

[1] Performance SLAs are applicable for data in hot tier
[2] Max allowed data in hot tier: 25%

I/O density calculations at the volume level are reported to show peak I/O density during the prior week. The peak performance is determined on an hourly time interval. I/O density reports by volume are generated monthly to gauge adherence to the respective service levels.

Service Level metrics and definitions

The following terms and definitions are used within the NetApp Keystone Flex Subscription (Flex subscription) service:

- **GiB, TiB, and PiB.** Measurements of data storage capacity using base of 1024 (1 GiB = 1024^3 bytes, 1 TiB = 1024^4 bytes, and 1PiB = 1024^5 bytes).
- **IOPS/TiB.** The protocol operations per second requested by the application divided by the allocated logical size of the volume.
- **Availability** is measured as a percentage of number of I/O requests successfully responded to by the service, divided by total number of I/O requests made of the service, measured at the service demarcation, in a given month, not including scheduled service downtime or unavailability of required facilities, network or other services to be provided by customer.
- **Durability** is the percentage of data accessed without loss of fidelity, excluding customer-caused deletion or corruption.
- **Target IOPS per TiB.** The guaranteed IOPS for all I/O requests made to a volume before the target IOPs per TiB threshold is reached. Performance on the volume is capped at the selected IOPS per TiB.

> The target IOPS per TiB performance metric is calculated based on the logical consumed capacity in TiB.

- **Latency.** Time to service an I/O request received from a client, measured at the service demarcation (storage controller I/O port).

Data Protection

NetApp Keystone Flex Subscription (Flex Subscription) data protection service can back up your data and is able to recover it if required. The available data protection services are:

- Snapshots of disks and shares
- Backups of disks and shares (requires data protection service as part of the subscription)
- Disaster recovery for disks and shares (requires data protection service as a part of the subscription)

> Backup and disaster recovery services are available as add-on services, while Snapshot is available as a part of the basic storage service.

	Single Region Snapshots (Available as a part of the basic storage service)	**Multi-region Backup (data protection add-on)**	**Multi-region Disaster Recovery (data protection add-on)**
Use case	Mitigate the risk of user or application data deletion or corruption, not against infrastructure loss or failure	Mitigate the risk of complete loss of data on the primary volume due to infrastructure loss or failure	Mitigate the risk of complete loss of data on the primary volume due to infrastructure loss or failure with a recovery time objective
Policy	Hourly, daily, weekly, and monthly	Number of backups to retain based on hourly, daily, weekly, and monthly Snapshots	1 hour, 4 hours, and daily
Topology	Source only	Backup	Async replication target

	Single Region Snapshots (Available as a part of the basic storage service)	Multi-region Backup (data protection add-on)	Multi-region Disaster Recovery (data protection add-on)
Target replication service level[1]	n/a	Standard	Same as primary

[1]Additional storage capacity to be subscribed

> Subscription to a basic Flex Subscription service does not automatically back up your data. You should subscribe to add-on data protection services and configure your system for data backup and disaster recovery services. If your storage system is not managed by Flex Subscription services, NetApp can still support protecting the data on your storage system and help in connecting it with your Flex Subscription services. However, NetApp is not responsible for any backup failures.

Tiering

NetApp Keystone Flex Subscription service includes a tiering capability that identifies less frequently used data and tiers it to a cold storage that is owned, deployed, and managed by NetApp.

The tiering capability leverages the NetApp FabricPool technology that enables automated tiering of data to low-cost object storage tiers either on or off premises. With this capability, infrequently accessed data is automatically tiered to a lower cost storage either on premises or in the cloud, based on the services agreed upon.

Partners and tenants can avail this capability easily by opting for the two preconfigured service levels, the Extreme-tiering and Premium-tiering service levels while provisioning their storage. The Extreme-tiering has the same QoS policies as the Standard, Extreme, and Premium service levels.

The add-on tiering capability is available only with Extreme and Premium service tiers. NetApp assumes 25% of data is hot and 75% is less frequently used and can be moved to a cold storage. Billing is determined based on the duration per volume is in each service level.

The following features are enabled:

- You can create reports of the inactive data for your disks and file shares and decide upon whether to change the service level. On moving or changing the tiering policy, the latencies can be higher if data is accessed from cold tier.
- You can change the service level of the volumes from Extreme and Premium to Extreme-tiering and Premium tiering respectively, provided that the destination tiering is enabled on the cluster.
- Likewise, you can change the tiering service levels to non-tiering for your volumes.
- Enable and disable backups for a volume on a tiering service level.
- Enable and disable disaster recovery for a volume on a tiering service level.

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